

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: Phase Ib Study of the Safety of T-DXd and Immunotherapy Agents With and Without Chemotherapy in Advanced or Metastatic HER2+, Non-squamous NSCLC **Official Title:** A Phase Ib Multicenter, Open-label Study to Evaluate the Safety and Tolerability of Trastuzumab Deruxtecan (T-DXd) and Immunotherapy Agents With and Without Chemotherapy Agents in First-line Treatment of Patients With Advanced or Metastatic Non-squamous Non-small Cell Lung Cancer (NSCLC) and Human Epidermal Growth Factor Receptor 2 (HER2) Overexpression (OE) (DESTINY-Lung03)
Short Title/Acronym: DESTINY-Lung03 (DL03) **Protocol Number:** D967YC00001 **NCT Number:** NCT04686305 **Posting ID:** 59943140253525160
Sponsor/Company Name: AstraZeneca **Trial Status:** Recruiting
Investigational Product: Trastuzumab deruxtecan (T-DXd / Enhertu), Durvalumab, Volrustomig (MEDI5752), Rilvegostomig, AZD2936, Pemetrexed, Cisplatin (Trade Name: Platinol, ATC Code: L01XA01), Carboplatin **Phase of Development:** Phase 1 (Phase 1b)
Medical Monitor: AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and dose-limiting toxicities (DLTs) of trastuzumab deruxtecan (T-DXd) in combination with immunotherapy agents, with and without chemotherapy, in patients with advanced or metastatic HER2-overexpressing non-squamous non-small cell lung cancer (NSCLC). **Secondary Objectives:** To evaluate the efficacy of these combination regimens, focusing on Confirmed Objective Response Rate (ORR), Duration of Response (DoR), and Disease Control Rate (DCR).

2.2 Background & Rationale To address the critical unmet medical need for effective HER2-directed treatment options in patients with locally advanced or metastatic HER2-overexpressing (HER2-OE) NSCLC. While T-DXd monotherapy has demonstrated activity in pretreated HER2-mutant NSCLC, this study explores whether combination therapies involving T-DXd alongside immunotherapy (durvalumab, volrustomig, rilvegostomig) and/or platinum-based chemotherapy can establish safe and effective novel regimens for both pretreated and treatment-naïve HER2-OE NSCLC populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Clinical Trial)
Control Type: Active/Uncontrolled (Multi-arm, dose-escalation and

dose-expansion) **Blinding:** Open-label **Randomization:** Randomized (Parallel group assignments for Parts 3, 4, and 5)

3.2 Study Parts & Cohorts * **Part 1 (Closed):** Dose escalation of T-DXd + durvalumab + cisplatin, carboplatin, or pemetrexed in pretreated patients. * **Part 2:** Cancelled (will not be initiated). * **Part 3 (Closed to recruitment):** T-DXd + volrustomig, without carboplatin (Arm 3A) or with carboplatin (Arm 3B). * **Part 4:** T-DXd + rilvegostomig, without carboplatin (Arm 4A) or with carboplatin (Arm 4B). Safety run-in followed by dose expansion. * **Part 5:** T-DXd + volrustomig evaluating priming vs. flat dosing regimens (Arm 5A, optional Arm 5B).

3.3 Planned Enrollment & Duration Target **NS:** 244 evaluable patients **Number of Sites:** Multicenter **Estimated Study Start:** 09-Mar-2021 **Estimated Primary Completion:** 30-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years, with WHO/ECOG performance status of 0 or 1, and minimum body weight of 35 kg (for Parts 3, 4, 5). 2. Histologically documented unresectable locally advanced/metastatic non-squamous NSCLC with HER2 overexpression status determined by central review. 3. Measurable target disease assessed by the investigator using RECIST 1.1. 4. Protocol-defined adequate organ and bone marrow function. 5. **Part 1:** Progression after 1 or 2 lines of systemic therapy for recurrent/metastatic disease. 6. **Parts 3, 4, and 5:** Treatment-naïve for advanced/metastatic NSCLC. Patients who received prior adjuvant/neoadjuvant chemo or definitive chemoradiation are eligible if progression occurred ≥ 6 months from the end of the last therapy. Tumors must not harbor known genomic alterations or actionable driver kinases for which approved therapies are available.

4.2 Exclusion Criteria 1. Known HER2 mutation. 2. History of (non-infectious) interstitial lung disease (ILD)/pneumonitis that required steroids, current ILD/pneumonitis, or suspected ILD/pneumonitis that cannot be ruled out by imaging. 3. Lung-specific intercurrent clinically significant illnesses (e.g., prior pneumonectomy, underlying pulmonary disorder). 4. Uncontrolled or significant cardiovascular disease, including myocardial infarction within 6 months, symptomatic CHF (NYHA II-IV), clinically important arrhythmias, recent stroke (< 6 months), cardiomyopathy, or troponin $> \text{ULN}$ at screening. History of thromboembolic events (PE, DVT, CVST) within 3 months. 5. Spinal cord compression or clinically active/symptomatic central nervous system (CNS) metastases requiring therapy. 6. Active primary immunodeficiency, known HIV infection, active/chronic Hep B or Hep C, active TB, or uncontrolled infection requiring IV antimicrobials. 7. Ascites or pericardial effusion requiring drainage, peritoneal shunt, or CART. 8. Active non-infectious

skin disease or autoimmune/inflammatory disorders requiring systemic/steroid treatment. 9. Unresolved toxicities not yet resolved to Grade ≤ 1 or baseline. 10. Medical contraindication to platinum-based chemotherapy. 11. Prior exposure to anti-PD-1, anti-PD-L1, anti-CTLA-4, anti-TIGIT, or other experimental immunotherapy (Parts 3, 4, 5). 12. History of substance abuse or psychological condition interfering with study participation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: T-DXd administered in varying combinations with immunotherapy (durvalumab, volrustomig, rilvegostomig) and/or chemotherapy (cisplatin [Platinol], carboplatin, pemetrexed). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Continued until objective disease progression, unacceptable toxicity, withdrawal of consent, or other discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy Permitted: Routine concomitant therapies for underlying comorbidities not interfering with IP. **Prohibited:** Concurrent anticancer treatments, live virus vaccines, chronic systemic corticosteroids, or other immunosuppressive therapies for autoimmune diseases.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy (Central HER2 review), Baseline Imaging, Virology (HIV, Hep B/C)
Baseline	Day 0	Randomization, First Dose Administration, Safety Baseline
Interim	Every 3 Weeks	Safety Labs, Efficacy Assessment (Tumor Imaging per RECIST 1.1), AE/SAE monitoring
End of Study		Follow-up Final Vital Signs, Exit Interview, Safety and survival tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Frequency of AEs and SAEs. **Measurement Method:** Occurrence of Adverse Events and Serious Adverse Events graded according to NCI CTCAE v5.0.

7.2 Secondary Endpoints 1. Confirmed Objective Response Rate (ORR). 2. Duration of Response (DoR). 3. Disease Control Rate (DCR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation of AEs during clinical visits, with specific rigorous monitoring for potential Interstitial Lung Disease (ILD) or pneumonitis via imaging. **Serious Adverse Events (SAEs):** Must be reported continuously throughout the evaluation window. **Data Monitoring Committee (DMC):** An internal Safety Review Committee (SRC) convenes during dose-escalation to evaluate DLTs and confirm recommended doses before dose-expansion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all patients receiving ≥ 1 dose of study drug); Efficacy Analysis Set (all treated patients with measurable disease). **Sample Size Justification:** Target enrollment is $N=244$. The dose-escalation/safety-run-in cohorts require 6 DLT-evaluable patients per arm. Dose-expansion cohorts enroll up to 34 patients per arm to adequately assess safety profiles and early efficacy signals. **Handling of Missing Data:** Standard survival analysis methods for time-to-event endpoints; right-censoring rules applied for patients without progression at data cutoff.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of participating centers to ensure compliance with trial protocols, particularly regarding complex dose-escalation and safety reporting rules. **Audit Trail:** Secure EDC (Electronic Data Capture) with thorough audit trails, mandatory central review of baseline tumor tissues, and Central Laboratory confirmation of HER2 status.

11. Study Identifiers & Governance

Primary ID: D967YC00001 **Registry Number:** NCT04686305 **Posting ID:** 59943140253525160 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** DESTINY-Lung03 (DL03) **Official Regulatory Title:** A Phase Ib Multicenter, Open-label Study to Evaluate the Safety and Tolerability of Trastuzumab Deruxtecan (T-DXd) and Immunotherapy Agents With and Without Chemotherapy Agents in First-line Treatment of Patients With Advanced or Metastatic Non-squamous Non-small Cell Lung Cancer (NSCLC) and Human Epidermal Growth Factor Receptor 2 (HER2) Overexpression (OE)

12. Clinical Framework (NSCLC Specific)

Primary Condition: Locally Advanced or Metastatic Non-Small Cell Lung Cancer **Preferred UMLS Name:** HER2-positive Non-Small Cell Lung Carcinoma **MeSH Code:** D002289 **HPO Code:** HP:0030358 **SNOMED Code:** 254637007 **Diagnosis Threshold:** HER2 overexpression status as determined by central review of tumor tissue **Medical Methodology:** Targeted Antibody-Drug Conjugate (ADC) combined with Immunotherapy and/or Platinum-based Chemotherapy

13. Study Procedures & Methodology

Management Pathway: Standard oncology pathway for advanced/metastatic NSCLC clinical trials **Education Protocol (HCP):** Site-specific training on dose-limiting toxicity (DLT) identification and stringent monitoring/management guidelines for ADC-related Interstitial Lung Disease (ILD) **Education Protocol (Patient):** Informed consent outlining trial phases (dose-escalation vs. expansion) and immediate self-reporting guidelines for respiratory symptoms **Quality Audit Mechanism:** RECIST 1.1 compliance review and Central Laboratory confirmation of HER2 status

14. Observation & Follow-Up Schedule

Total Duration: Treatment until disease progression; Follow-up for post-treatment safety and survival **Onsite Visit Cadence:** Every 3 weeks (corresponding to the standard T-DXd/Immunotherapy infusion cycles) **Remote Monitoring Frequency:** Continuous safety and survival tracking post-treatment discontinuation **Checklist Review Interval:** Scheduled radiological imaging for tumor response assessment per RECIST 1.1

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				